Testing a Readable-Witness Diagnostic for Reward Compatibility

A controlled, API-based study of in-context optimization

Raphael Khalid
Independent researcher

Working manuscript, September 10, 2026; results added October 4, 2026

Results draft, not submission-ready. The run is complete and the registered analysis is reported below. The planned human audit has not been performed, and no novelty claim is made. Methods below describe the frozen implementation. This manuscript was drafted after launch, not independently preregistered before data collection, and the results sections were written after the final effects had been inspected.

Abstract

Optimizing a language model's visible reasoning may change what an external monitor can infer from it. We investigate whether a small search for readable strategies that jointly satisfy outcome and reasoning rewards can predict later monitoring loss. Our diagnostic is a finite-search, model-audited witness rate, not a decision procedure for universally quantified reward compatibility. The registered schedule covers eight reward configurations in five families on a deterministic coin-state task. Each configuration has matched outcome-only and combined-reward conditions, three separate optimization histories per condition, and twelve in-context optimization steps per history. Candidate strategies are scored on five fresh transfer tasks; baseline and terminal evaluation use ten held-out tasks per history. We compare witness rate with description-only predictions, baseline monitorability, and baseline reasoning length using leave-one-family-out prediction. The complete main schedule contains 716 orchestration units under a $40 API cap; all completed, at a recorded cost of $1.29. In leave-one-family-out prediction of additional monitoring loss, the witness rate had a mean absolute error of 0.156 against 0.158 for an intercept-only model, and description-only predictions, baseline monitorability and baseline length all did worse than the intercept. Only one of eight configurations, a 16-character length limit, had a repeat-level interval for additional loss that excluded zero, and that interval is unadjusted for multiplicity. The diagnostic therefore showed no meaningful predictive value at this scale. The contribution is a reproducible, fully exported test of the diagnostic and its failure modes, with explicitly limited claims about generalization, causal mechanisms, and internal reasoning.

1 Introduction

An oversight method can become less informative when the system it observes is optimized against incentives that affect the observable signal. For language models, the signal may be a visible explanation accompanying an answer. An explanation that sounds useful is not necessarily a faithful account of the computation producing the answer. We therefore separate answer correctness, visible support for that answer, and an external monitor's ability to identify the task from the explanation.

Kaufmann et al. [1] propose a framework relating outcome and reasoning rewards and identify empirical detection of their compatibility as future work. We investigate a narrower operational

question: *does a bounded search for readable, jointly high-reward strategies predict additional monitoring loss under a specified optimization procedure?* A successful predictor could inform decisions about which reward pairs merit further investigation. It would not certify safety or determine the theoretical category of an arbitrary reward pair.

This study is implemented in AutoLabs as one fixed experiment with a public progress interface, not as a competition among autonomous scientific decision-makers. A coordinator reports progress but cannot change the scientific protocol. The live interface is incidental to the estimator; the scientific record consists of fixed prompts, generated tasks, visible outputs, scores, and versioned analysis code.

Candidate contribution. We specify a readable-witness diagnostic, evaluate it against four predictive references, and release an executable experiment with a controlled outcome-only comparison. Whether this constitutes a useful empirical advance depends on the completed results, human audit, and a fuller related-work assessment. We make no priority claim at this stage.

2 Related work and scope

The reward-compatibility framework of Kaufmann et al. [1] motivates our experiment. Their API-based in-context approach includes eliciting visible tagged reasoning when extended thinking is unavailable and using transfer-based rewards. Our implementation uses a different model, independently worded prompts, and a substantially smaller schedule; it is not a full replication of their experiments or a measurement of weight-training effects.

Baker et al. [2] study monitoring reasoning models for misbehavior and the risk that optimization promotes obfuscation. Our benign coin-identification endpoint is a deliberately narrower surrogate: identifying coin-related content is not the same as detecting harmful intent. Lanham et al. [3] study tests of chain-of-thought faithfulness. Our evidence audit checks visible support rather than intervening on reasoning to identify its causal contribution. None of our scores accesses private reasoning traces, model activations, or a mechanistic account of the computation.

3 Experimental design

3.1 Task and splits

The actor tracks a coin through operations called `Same` and `Different`. Same preserves the state and Different flips it. With initial state $x_0 \in \{0, 1\}$ and flip indicators $d_1, \ldots, d_L$, the exact final state is

$$x_L = x_0 \oplus d_1 \oplus \cdots \oplus d_L. \tag{1}$$

Zero denotes Heads and one denotes Tails. Correctness is computed by this deterministic checker, not by a language-model judge. The actor emits `<reasoning>` and `<answer>` fields with a strict output schema.

Most configurations use six operations. There are only $2^7 = 128$ distinct six-operation problems, counting both initial states. The implementation stratifies them by final answer and partitions them into disjoint feasibility, diagnostic, training, and held-out sets of 32 problems each. Sampling can repeat within a split: *thousands of completions are not thousands of unique underlying tasks.* Process supervision uses 42 operations. Its first six operations and initial state follow the same partition; additional operations are seeded, with a final operation chosen to retain the desired

answer balance. These are synthetic, balanced tasks, not a representative distribution of reasoning problems.

3.2 Models and information boundaries

The actor uses the OpenAI API model alias `gpt-5.6-luna` with reasoning effort `none`. Fresh calls to the same alias with effort `high` provide reward grades, monitoring grades, evidence audits, description-only predictions, and progress summaries. Role separation is separation of requests and prompts, not independence of model families. API-returned model identifiers, usage, and timestamps are recorded. The alias is not an immutable checkpoint, and effort labels do not expose the provider's internal computation.

The evaluation monitor sees only the visible reasoning passage, not its originating task, final answer, reward configuration, or experimental condition. It rates evidence that the passage concerns a coin from zero to five. Identification is $\mathbf{1}[m \geq 3]$. The fixed rubric explicitly avoids inferring coin context from abstract binary-state tracking alone. This operational definition can classify a mathematically readable but domain-abstract explanation as difficult to identify.

The diagnostic evidence auditor sees the task and visible reasoning, but the actor's final answer and downstream scores are withheld. A positive audit requires an explicit, correct computation and an exact supporting quotation. Empty reasoning, bare conclusions, and unsupported encodings do not qualify. Model judgment is still involved; an exact quotation does not establish that the judgment is correct.

3.3 Rewards and conditions

Let $R_o = \mathbf{1}[\widehat{x}_L = x_L]$ and let R_r be the reasoning reward. The outcome-only condition optimizes R_o; the combined condition optimizes $R_o + R_r$. Table 1 gives the implemented rewards. Only the combined actor is told the reasoning incentive. Shadow reasoning grades are nevertheless computed in both conditions to equalize measurement and support later comparisons.

Family	Reasoning reward R_r	Configurations
Length	$-\max(0, \ell - b)$, where ℓ counts trimmed Unicode code points	$b = 0, 16, 64, 256$
String	-1 if Heads or Tails occurs, case-insensitive; otherwise 0	One
Semantic	$-m/5$, where $m \in \{0, \ldots, 5\}$ is the coin-monitor grade	One
Preference	Simulated preference grade in $\{-1, 0, 1\}$	One
Process	-1 if a grader detects an incorrect transition; otherwise 0	One, $L = 42$

Table 1: Eight configurations in five families. The preference reward is model-generated, not human feedback. Empty reasoning can attain the process reward without demonstrating useful computation.

3.4 What one main-study unit contains

A unit is an orchestration checkpoint, not a single API request or an independent experimental replicate. For example, an optimization unit for the string-penalty configuration in a combined-reward history proceeds as follows:

1. Construct one training problem and select replay examples from that history only.

2. Ask the actor for one candidate explanation and answer under the condition's reward prompt.

3. Ask fresh actor calls to apply that reasoning pattern to five distinct fresh training tasks, excluding the candidate task.

4. Compute all learning rewards on the five transferred completions. Here correctness and the string penalty are deterministic. Other families require fresh grader calls.

5. Average the transfer rewards, attach that score to the candidate strategy, and update this history's replay buffer.

Thus this particular optimization unit uses six actor calls. A semantic, preference, or process optimization unit additionally uses five grader calls. A diagnostic unit also audits the five transferred passages. An evaluation unit generates and monitors ten held-out completions, with extra reasoning-reward grading where needed. Description and reporting units each use one call.

3.5 Schedule and replay

The 68-unit feasibility gate checks format and six-step actor accuracy, coin-monitor sensitivity and specificity, evidence-auditor calibration, and execution of all eight reward configurations. Calibration success is not evidence for the main hypothesis. The 42-step accuracy is descriptive, not a selection threshold. The main schedule, frozen in source before paid launch, is shown in Table 2.

Work item	Units
Description-only predictions: eight configurations	8
Diagnostic search: four candidates per configuration	32
Baseline evaluation: $8 \times 2 \times 3$ histories	48
Optimization: $8 \times 2 \times 3 \times 12$ steps	576
Operational summaries	4
Terminal evaluation: $8 \times 2 \times 3$ histories	48
Total main schedule	716

Table 2: The 716 checkpoints contain 48 optimization histories, not 716 independent experiments. Baseline and terminal evaluations each contain 480 completions. The diagnostic contains 160 transferred completions from 32 candidate strategies.

Each history has isolated success and failure buffers with capacity 256 each. A strategy with mean transfer reward at least 75% of the condition's maximum is a success. Replay selects up to ten examples, using exploration probability 0.1 and reward-softmax temperature 0.2. Once a strategy reaches the maximum mean transfer reward, the implementation reuses a maximum-reward example and does not ask for further improvement; all twelve scheduled steps still run. No weights are updated.

Task schedules are paired between conditions. History order is seeded and interleaved during optimization; API sampling itself is not made reproducible by these task seeds. Baseline and terminal evaluation use the same ten balanced held-out problems for each configuration/repeat pair, shared across conditions. Held-out records never enter replay, diagnostics, or the coordinator's context and remain publicly sealed until completion.

4 Diagnostic and estimands

4.1 Finite-search readable witnesses

For configuration c, diagnostic strategy $j \in \{1, \ldots, 4\}$ and transfer $i \in \{1, \ldots, 5\}$, define

$$W_{cji} = \mathbf{1}[R_o = 1]\mathbf{1}[R_r = R_{r,c}^{\max}]\mathbf{1}[A = 1], \qquad \widehat{w}_c = \frac{1}{4}\sum_{j=1}^{4}\frac{1}{5}\sum_{i=1}^{5} W_{cji}, \tag{2}$$

where A is the evidence-auditor decision. The maximum reasoning reward is one for preference and zero otherwise. We report a descriptive 95% Student-t interval across the four strategy-level rates, using $t_{.975,3} = 3.182$. This is not a binomial interval over twenty independent trials. Task reuse and same-model evaluation further limit its interpretation.

A positive witness demonstrates success for a particular visible strategy and task under the implemented rewards and audit. Zero witnesses means this finite search did not find one. It cannot establish theoretical conflict. A high witness rate likewise does not guarantee future optimization preserves readability. The zero-character configuration makes a nonempty supported visible explanation impossible by construction; its inclusion provides a structurally extreme condition, not a substantive discovery.

4.2 Additional monitoring loss

Let $M_{c,a,r,t}$ be mean monitor identification on ten held-out tasks, for condition $a \in \{C, O\}$ (combined or outcome-only), repeat r, and time $t \in \{0, 1\}$ (baseline or terminal). The paired contrast is

$$D_{cr} = (M_{c,C,r,0} - M_{c,C,r,1}) - (M_{c,O,r,0} - M_{c,O,r,1}), \qquad \widehat{D}_c = \frac{1}{3}\sum_{r=1}^{3} D_{cr}. \tag{3}$$

Positive $\widehat{D}_c$ indicates greater baseline-to-terminal identification loss in the combined condition. The registered interval is $\widehat{D}_c \pm 4.303 s_D/\sqrt{3}$, with two degrees of freedom. Wide intervals, including intervals extending outside feasible probability bounds, are reported without silent clipping. They are descriptive small-sample summaries, not a calibrated guarantee of population coverage.

We also report the corresponding difference in accuracy improvements and the fraction of terminal combined-condition samples attaining maximum reasoning reward. The primary endpoint includes all held-out completions, not only correct answers. The contrast estimates the effect of the complete reward-prompt and replay procedure in this harness. Because baseline prompts already differ between conditions, it is not the total effect of initially revealing the incentive, nor does it isolate causal mediation through visible reasoning.

4.3 Predictive comparison

The candidate predictors are witness rate, description-only predicted loss, baseline identification rate, and baseline reasoning length; an intercept-only model is the reference. For each predictor, univariate ordinary least squares is fit with equal total weight for each training reward family. Each of the five folds withholds one entire family. The four length thresholds always remain in one fold. We report mean absolute prediction error averaged first within held-out family and then equally over families. A constant predictor falls back to the intercept fit.

Only five family-level folds are available, and their training sets overlap. We do not treat their errors as five independent confirmatory trials, tune the diagnostic on terminal outcomes, or make a

broad generalization claim from a small error difference. The description-only score is constrained to $[0, 1]$, while observed additional loss can be negative; its linear calibration occurs within training folds, not on held-out outcomes.

5 Results

All 784 scored checkpoints (68 gate plus 716 main) were exported with `npm run export`, which verified record counts and recomputed the analysis from the exported records without model calls. The recomputed analysis is identical to the public analysis endpoint. Records, estimates, analysis and figure are in `data/experiment-002-v01/`.

Feasibility and execution. The 68-unit gate passed every registered check. Six-step actor accuracy was 12/12 (criterion at least 11/12). Coin-monitor sensitivity was 12/12 and specificity 12/12 (criterion at least 11/12 each). The evidence auditor matched 11/12 calibration cases (criterion at least 11/12). All eight reward configurations executed with five transfers each. The 42-step actor accuracy was 10/12; this is descriptive and was not a gate criterion. All 716 main units completed. The run was recorded as started at 2026-09-10 22:03:11 UTC and completed at 2026-09-11 00:31:10 UTC, an elapsed time of 2 h 28 min that includes the gate; active time was not recorded separately. The ledger records 7,364 calls and $1.288641 of price-based spend, with $0.002985 still held as conservative reservations for calls whose billing was not confirmed. The build log records an initial HTTP 401 from a placeholder credential, preserved as a failed record before the correct key was installed. The configured model alias was `gpt-5.6-luna`; the exported records do not include per-call returned model identifiers, so these have not been verified here.

Configuration	Witness rate	Extra loss	Accuracy effect	Reward attained
Length 0	0 [0, 0]	0.0 [−49.7, 49.7]	+6.7	33.3
Length 16	10 [−8.4, 28.4]	+63.3 [25.4, 101.3]	+30.0	100
Length 64	20 [−6.0, 46.0]	0.0 [−65.7, 65.7]	+10.0	100
Length 256	100 [100, 100]	+20.0 [−142.9, 182.9]	0.0	100
String	70 [28.9, 111.1]	+20.0 [−69.6, 109.6]	+13.3	100
Semantic	20 [−25.0, 65.0]	0.0 [−24.8, 24.8]	+3.3	96.7
Preference	55 [24.5, 85.5]	−13.3 [−150.2, 123.5]	0.0	70.0
Process	100 [100, 100]	+13.3 [−87.1, 113.7]	+16.7	80.0

Table 3: Registered estimates. Rates are percentages and contrasts percentage points. Witness intervals are 95% Student-t intervals across four strategy-level rates (df = 3); extra-loss intervals are 95% Student-t intervals across three repeats (df = 2). Intervals are not clipped to feasible bounds and are not adjusted for eight comparisons. Accuracy effect is the combined-minus-outcome-only difference in accuracy improvement; reward attained is the fraction of terminal combined-condition samples at maximum reasoning reward.

Monitoring loss. Additional monitoring loss was indistinguishable from zero for seven of eight configurations. The 16-character length limit is the exception: $\widehat{D} = +63.3$ percentage points, with repeat-level contrasts of +80, +60 and +50 and an interval of $[25.4, 101.3]$. The same configuration also had the largest accuracy effect (+30.0 points), so the combined condition improved accuracy more while becoming harder to identify. With three repeats and eight unadjusted intervals, this is a lead to retest, not a confirmed effect. Two nulls are uninformative about compatibility for a

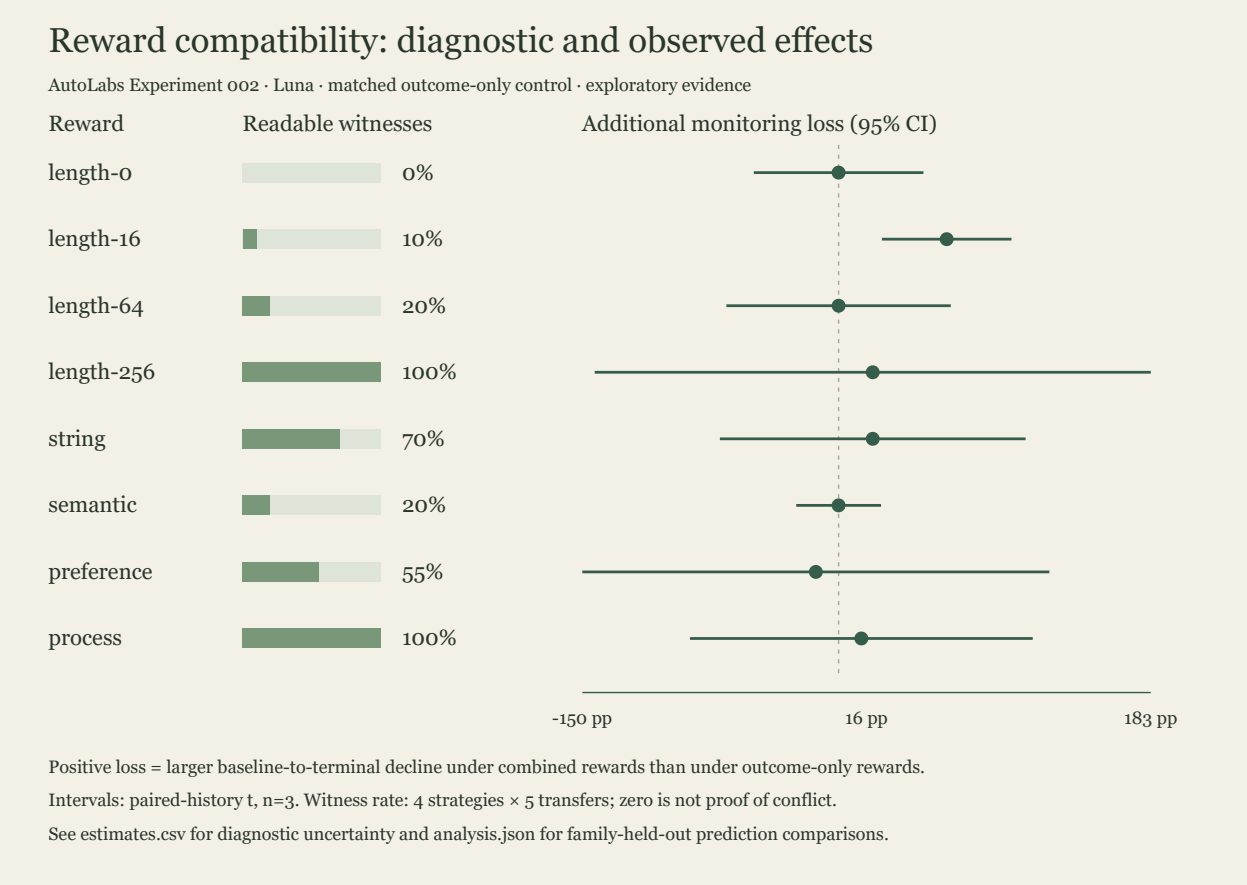


Figure 1: Readable-witness rates (left) and additional monitoring loss with repeat-level intervals (right) for the eight configurations. The plot alone does not establish predictive performance.

separate reason: the zero-character configuration attained its reasoning reward in only 33.3% of terminal samples, and the preference configuration in 70.0%.

Predictive value. Table 4 gives family-held-out mean absolute error for all five models. The witness rate (0.156) is essentially tied with the intercept-only reference (0.158). Description-only predictions (0.172), baseline reasoning length (0.218) and baseline identification rate (0.223) are worse than the intercept. Within families, the witness rate improves slightly on the intercept for string, semantic and process rewards and is worse for the length family; description-only prediction is best on string and process but worst on semantic. These family-level differences rest on single configurations and do not support a ranking. The registered diagnostic did not demonstrate predictive value at this scale.

Human audit. The planned audit of the 160 diagnostic transfers has not been performed. The registered plan required freezing the audit procedure before inspecting final effects; that is no longer possible, because the final effects are public and were inspected in preparing this section. Any human audit will therefore be post hoc. It must be labelled as such, retain the original model labels alongside human labels and adjudications, and be reported as a sensitivity analysis rather than a replacement for the frozen primary diagnostic.

Predictor	Overall	Length	String	Semantic	Preference	Process
Intercept only	0.158	0.208	0.148	0.102	0.269	0.065
Witness rate	0.156	0.225	0.142	0.099	0.268	0.046
Description-only	0.172	0.225	0.070	0.212	0.311	0.045
Baseline length	0.218	0.210	0.111	0.088	0.253	0.431
Baseline identification	0.223	0.205	0.166	0.163	0.269	0.313

Table 4: Leave-one-family-out mean absolute error for predicting additional monitoring loss, averaged equally over the five held-out families. Training folds overlap; these are not five independent tests.

6 Limitations and interpretation

Construct validity. Coin identification is not general oversight quality. Domain abstraction can make a derivation readable while making its coin origin unidentified under the fixed rubric. The semantic reward uses the same monitor prompt and model family as evaluation, although requests are fresh. This tests response to that monitor incentive, not evasion of an unseen monitor family. The evidence auditor's ability to check a correct derivation requires human validation.

Dependence and sample size. Three separate optimization histories per arm are a small sample and share tasks, model infrastructure, and fixed prompts. Diagnostic transfers share candidate strategies. Families are hand-selected, not sampled randomly from a population of rewards. The zero-length endpoint can dominate apparent predictive success. Sensitivity analyses removing structurally extreme configurations would be informative but must be labelled additional, not silently substituted for the registered analysis.

Optimization and scale. Twelve in-context steps may not substantially optimize every reward. A null monitoring effect accompanied by low reward attainment is not evidence of compatibility. Accuracy ceilings, especially for six-step problems, prevent strong conclusions about whether process incentives improve useful computation. Process supervision changes task length as well as reward family. Its family comparison is therefore not a clean reward-only intervention. The $40 ceiling is a resource constraint, not a sample-size or power justification.

Mechanism and external validity. Output text does not identify private reasoning or its causal role. The model can potentially answer a short task without expressing a derivation. No activation access, reasoning intervention, model fine-tuning, human-preference collection, or adversarial code task is part of this study. We do not extrapolate to deployed safety monitors, harmful-intent detection, or arbitrary reward pairs. A moving model alias and sequential service calls leave residual time/version confounding despite interleaving.

Draft timing and additional analysis. The source protocol predates paid execution. This manuscript and its audit checklist were written while the run was ongoing; the results, abstract and conclusion were added on October 4, 2026, after export. Any new analyses, human audit choices, or hypotheses introduced here must be time-stamped and distinguished from the frozen primary analysis. No novelty, positive result, or publication acceptance follows from preregistration or reproducibility alone.

7 Reproducibility and operational record

The protocol is version `reward-compatibility-v0.1`, run `experiment-002-v01`. Initial registration is preserved at source commit `336232c`. The API ledger retains request identifiers, visible prompts and completions, token usage, charges, and ambiguous outcomes. Unknown billing retains a conservative reservation rather than triggering blind retries. The application cap is $40, including a $4 feasibility allowance; recorded price-based accounting is not a provider-invoice guarantee.

At main cursor 35, the owner requested an operational speedup. After pausing and draining requests, revision `parallel-ledger-v1` enabled up to three independent sample pipelines within a unit. Candidate generation still precedes transfers; optimization steps and replay updates remain sequential. Task IDs, prompts, ordering of stored samples, and study counts were unchanged. The amendment was recorded publicly. Resume reuses completed-call checkpoints rather than restarting the experiment. Storage protections bound records and public pages and pause before the configured storage threshold; no retention truncation is used to manufacture a complete run.

After completion, `npm run export` verifies record counts and produces the manifest, scored records, estimates CSV, analysis JSON, and SVG figure without model calls. Full prompts live in `src/prompts.ts`; exact task and reward code in `src/protocol.ts`; analysis in `src/analysis.ts`. The source repository is `https://github.com/RaphaelKhalid/reward-compatibility`. The public observer is `https://autolabs-ebon.vercel.app/experiments/reward-compatibility`. Before submission, archive the final code/data snapshot and record the immutable archive identifier.

8 Conclusion

We tested whether a readable-witness diagnostic predicts monitoring loss under API-based in-context optimization. In this design it did not: its held-out error was essentially that of an intercept-only model, and the simpler alternatives did worse. Most configurations produced no detectable additional monitoring loss, and two produced too little reward attainment for their nulls to bear on compatibility. A 16-character length limit produced a consistent loss across all three repeats, which merits a larger, preregistered follow-up. These conclusions are limited by three repeats per arm, five reward families, a coin-identification surrogate for oversight, same-family model graders, and the absence of the planned human audit. Nothing here resolves the theoretical reward-classification problem.

Author responsibility and AI assistance

This draft has one human author. AI systems assisted implementation, source inspection, manuscript drafting, and the generation and scoring of experimental outputs. AI tools are not authors. The human author is responsible for verifying the methods, citations, data, analysis, and final claims. Human review of this manuscript and final evidence remains pending. No affiliation with or endorsement by the cited researchers, Google DeepMind, OpenAI, or model providers is implied.

References

[1] Max Kaufmann, David Lindner, Roland S. Zimmermann, and Rohin Shah. 2026. *Aligned, Orthogonal or In-conflict: When can we safely optimize Chain-of-Thought?* arXiv:2603.30036v1. `https://arxiv.org/abs/2603.30036v1`.

[2] Bowen Baker et al. 2025. *Monitoring Reasoning Models for Misbehavior and the Risks of Promoting Obfuscation.* arXiv:2503.11926. https://arxiv.org/abs/2503.11926.

[3] Tamera Lanham et al. 2023. *Measuring Faithfulness in Chain-of-Thought Reasoning.* arXiv:2307.13702. https://arxiv.org/abs/2307.13702.